

Bingqin Wang

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Education

University of Illinois Urbana-Champaign

Aug 2026 – May 2028 (expected)

Master of Computer Science

Southern University of Science and Technology

Sep 2022 – Jul 2026

Bachelor of Engineering in Computer Science and Engineering

Experience

State Labs (Lingtai Technology)

May 2026 – Aug 2026

Software Engineer Intern

- Developed a Spring Boot API gateway exposing generation APIs compatible with OpenAI, Gemini, and Anthropic clients and returning JSON responses or SSE streams.
- Implemented a Gemini adapter that maps conversation history and generation parameters to an OpenAI-compatible upstream and translates responses and SSE chunks back into Gemini format.
- Implemented optimistic locking and bounded retries for balance deductions, recorded partial charges, stripped client-supplied identity headers, and added JUnit and Mockito tests for billing logic.

Stony Brook University

Apr 2025 – Nov 2025

Research Assistant, AI Innovation Institute

- Built a PyTorch pipeline for weakly supervised segmentation, generating pixel-level pseudo-labels from image-level tags.
- Trained segmentation models with ResNet-50 and DINOv3 backbones using multi-GPU mixed precision, reaching 76.8% mIoU with DINOv3 on VOC.

Projects

Distributed Workflow Engine for GitHub Analytics

Mar 2026 – May 2026

Python, PostgreSQL, asyncio, Docker

- Built a distributed DAG engine for GitHub activity analytics, coordinating schedulers and workers through PostgreSQL transactions, task leases, and heartbeats.
- Implemented dynamic task expansion for parallel data processing and asynchronous triggers that release worker slots while waiting for external data.
- Added partitioned backfills and idempotent result publication, using fault injection to verify output consistency across worker crashes, retries, and repeated backfills.

Real-Time Collaborative Markdown Editor

Oct 2025 – Dec 2025

Go, TypeScript, PostgreSQL, WebSocket, IndexedDB

- Built an operational transformation engine in Go and TypeScript for collaborative Markdown editing, validating document convergence under concurrent edits with shared cross-language tests.
- Designed a WebSocket synchronization protocol with operation IDs, acknowledgments, and durable deduplication, and persisted pending edits in IndexedDB for recovery after disconnects and page reloads.
- Implemented transactional operation logs and versioned snapshots in PostgreSQL, acknowledging edits only after commit and testing recovery from crashes before and after transaction completion.

Technical Skills

Languages: Java, Python, Go, C++, TypeScript, JavaScript, SQL, Bash

Backend: Spring Boot, Spring Cloud Gateway, FastAPI, REST APIs, PostgreSQL, MySQL, Redis, MyBatis, Kafka

Frontend: React, TypeScript, HTML, CSS

DevOps: Docker, Linux, Nginx, Git, GitHub Actions, CI/CD, Maven

CS Fundamentals: distributed systems, concurrency, operating systems, networking, security, object-oriented design